

11 **(short)** A tow truck of mass M is connected to a small car of mass m by a stretchable elastic rope, which is modelled with a viscous dashpot λ and a spring k , as shown in Fig. 11.

(a) The truck and car are both stationary, and the rope is at its natural length. At time $t = 0$, the truck moves away from the car at constant velocity v . If the position of the car is x , show that the equation of motion for x is

$$m\ddot{x} = k(vt - x) + \lambda(v - \dot{x}). \quad [4]$$

(b) The viscous dashpot is tuned so the system is critically damped. Assuming $x = 0$ at time $t = 0$, find the motion of the car $x(t)$. [6]

Hint: You may assume that the solution to the critically damped equation of motion $\ddot{y}/\omega_n^2 + 2\dot{y}/\omega_n + y = 0$ is of the form $y = (At + B)e^{-\omega_n t}$.

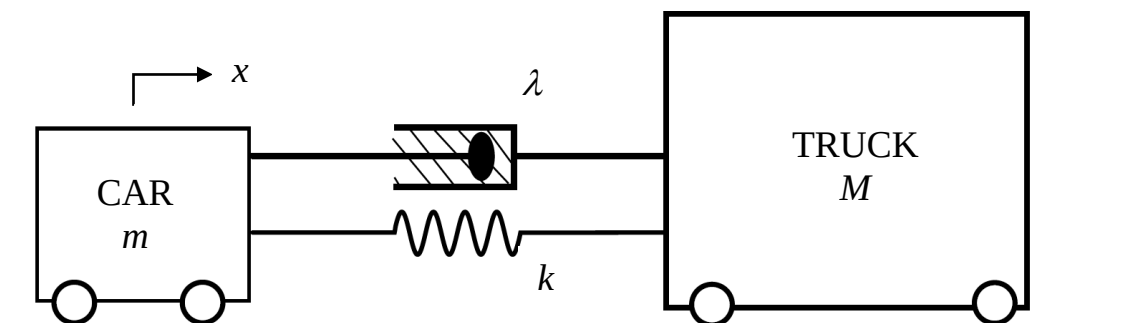


Fig. 11